

Assignment 6 Probability

Each complete question and its options are followed by the correct answer.

Question 1: The probability of an event can have a value between: A. -1 and 1 B. 0 and 1 C. 1 and 100 D. $-\infty$ and $+\infty$

Answer: B. 0 and 1

Question 2: If the probability of an event is 0, the event is: A. Certain B. Likely C. Impossible D. Equally likely

Answer: C. Impossible

Question 3: If the probability of an event is 1, the event is: A. Impossible B. Certain C. Unlikely D. Random

Answer: B. Certain

Question 4: A fair coin is tossed once. What is the probability of getting Heads? A. 0 B. 0.25 C. 0.5 D. 1

Answer: C. 0.5

Question 5: A fair six-sided die is rolled once. What is the probability of getting a 4? A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{1}{6}$ D. $\frac{4}{6}$

Answer: C. $\frac{1}{6}$

Question 6: A fair die is rolled once. What is the probability of getting an even number? A. $\frac{1}{6}$ B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{2}{3}$

Answer: C. $\frac{1}{2}$

Question 7: If $P(A) = 0.3$, what is $P(\text{not } A)$? A. 0.3 B. 0.5 C. 0.7 D. 1.3

Answer: C. 0.7

Question 8: Which of the following represents the complement of event A? A. $A + B$ B. $A \cap B$ C. $A \cup B$ D. A'

Answer: D. A'

Question 9: A sample space represents: A. Only successful outcomes B. All possible outcomes of an experiment C. Only failed outcomes D. The probability of an event

Answer: B. All possible outcomes of an experiment

Question 10: When a die is rolled once, the sample space is: A. {0, 1, 2, 3, 4, 5} B. {1, 2, 3, 4, 5, 6} C. {2, 4, 6} D. {1, 3, 5}

Answer: B. {1, 2, 3, 4, 5, 6}

Question 11: If A represents getting an even number when rolling a die, which set represents A? A. {1, 3, 5} B. {2, 4, 6} C. {1, 2, 3} D. {4, 5, 6}

Answer: B. {2, 4, 6}

Question 12: Two events are mutually exclusive when: A. They always occur together B. They cannot occur at the same time C. They have equal probability D. They have equal outcomes

Answer: B. They cannot occur at the same time

Question 13: Which pair of events is mutually exclusive when rolling a die? A. Getting an even number and getting a 4 B. Getting a number greater than 3 and getting a 6 C. Getting an odd number and getting an even number D. Getting a number less than 5 and getting a 3

Answer: C. Getting an odd number and getting an even number

Question 14: If two events A and B are mutually exclusive, which expression gives $P(A \text{ or } B)$? A. $P(A) \times P(B)$ B. $P(A) + P(B)$ C. $P(A) - P(B)$ D. $P(A) / P(B)$

Answer: B. $P(A) + P(B)$

Question 15: If $P(A) = 0.4$ and $P(B) = 0.3$, and A and B are mutually exclusive, what is $P(A \text{ or } B)$? A. 0.12 B. 0.1 C. 0.7 D. 1.2

Answer: C. 0.7

Question 16: If two events can occur at the same time, simply adding their probabilities may: A. Always give the correct answer B. Double -count their overlap C. Make the probability zero D. Make the events independent

Answer: B. Double -count their overlap

Question 17: Suppose: $P(A) = 0.5$ $P(B) = 0.4$ $P(A \text{ and } B) = 0.2$ What is $P(A \text{ or } B)$? A. 0.2 B. 0.7 C. 0.9 D. 1.1

Answer: B. 0.7

Question 18: Conditional probability answers which question? A. What is the probability of A given that B has occurred? B. What is the probability of A and B never occurring? C. What is the probability of all possible events? D. What is the average probability?

Answer: A. What is the probability of A given that B has occurred?

Question 19: The notation $P(A | B)$ means: A. Probability of A and B B. Probability of A given B C. Probability of B given A D. Probability of neither A nor B

Answer: B. Probability of A given B

Question 20: If $P(A \text{ and } B) = 0.2$ and $P(B) = 0.5$, what is $P(A | B)$? A. 0.1 B. 0.2 C. 0.4 D. 0.7

Answer: C. 0.4

Question 21: A medical test is performed only on patients who have already been identified as high -risk. The probability of a positive result in this group is an example of: A. Conditional probability B. Independent probability C. Complementary probability D. Uniform probability

Answer: A. Conditional probability

Question 22: If knowing that event B occurred changes the probability of event A, then A and B are generally: A. Independent B. Dependent C. Mutually exclusive D. Identical

Answer: B. Dependent

Question 23: Two events are independent when: A. They cannot occur together B. The occurrence of one does not affect the probability of the other C. They always occur together D. Their probabilities are equal

Answer: B. The occurrence of one does not affect the probability of the other

Question 24: A fair coin is tossed twice. Is the result of the second toss affected by the result of the first toss? A. Yes, always B. No C. Only when the first toss is Heads D. Only when the first toss is Tails

Answer: B. No

Question 25: If A and B are independent, which expression gives $P(A \text{ and } B)$? A. $P(A) + P(B)$ B. $P(A) - P(B)$ C. $P(A) \times P(B)$ D. $P(A) / P(B)$

Answer: C. $P(A) \times P(B)$

Question 26: If $P(A) = 0.5$ and $P(B) = 0.2$, and A and B are independent, what is $P(A \text{ and } B)$? A. 0.1 B. 0.3 C. 0.7 D. 1.0

Answer: A. 0.1

Question 27: Which pair of events is most likely to be independent? A. Drawing two cards from a deck without replacement B. Tossing a coin twice C. Selecting two students without replacement D. Taking two balls from a bag without replacement

Answer: B. Tossing a coin twice

Question 28: A fair die is rolled twice. How many possible ordered outcomes are there? A. 6 B. 12 C. 36 D. 42

Answer: C. 36

Question 29: A coin is tossed three times. How many possible outcomes are there? A. 3 B. 6 C. 8 D. 9

Answer: C. 8

Question 30: If there are 4 shirts and 3 pairs of trousers, how many different shirt -trouser combinations are possible? A. 7 B. 12 C. 16 D. 24

Answer: B. 12

Question 31: Which concept is used when the order of selection matters? A. Combination B. Permutation C. Mean D. Variance

Answer: B. Permutation

Question 32: Which concept is used when the order of selection does NOT matter? A. Permutation B. Combination C. Conditional probability D. Independence

Answer: B. Combination

Question 33: In how many ways can 3 different students be arranged in a row? A. 3 B. 6 C. 9 D. 12

Answer: B. 6

Question 34: A random variable is: A. A variable whose value is determined by the outcome of a random experiment B. A variable that must always be random in a computer program C. A variable that can only contain integers D. A variable with probability 1

Answer: A. A variable whose value is determined by the outcome of a random experiment

Question 35: Which of the following is an example of a discrete random variable? A. Height of a person B. Temperature C. Number of defective products D. Weight of a person

Answer: C. Number of defective products

Question 36: Which of the following is an example of a continuous random variable? A. Number of students B. Number of defective items C. Number of cars D. Temperature

Answer: D. Temperature

Question 37: The sum of probabilities of all possible outcomes of a discrete random variable should be: A. 0 B. 0.5 C. 1 D. Greater than 1

Answer: C. 1

Question 38: Expected value can be interpreted as: A. The most frequent outcome in every experiment B. The long -run weighted average outcome C. The maximum possible outcome D. The minimum possible outcome

Answer: B. The long -run weighted average outcome

Question 39: A random variable X has: $X = 0$ with probability 0.5 $X = 10$ with probability 0.5 What is its expected value? A. 2.5 B. 5 C. 10 D. 20

Answer: B. 5

Question 40: A game gives ■100 with probability 0.2 and ■0 with probability 0.8. What is the expected value? A. ■10 B. ■20 C. ■50 D. ■80

Answer: B. ■20

Question 41: Bayes' theorem is primarily concerned with: A. Updating probabilities based on new evidence B. Calculating averages C. Finding matrix rank D. Calculating variance

Answer: A. Updating probabilities based on new evidence

Question 42: Suppose a medical test is positive. You want to know the probability that the patient actually has the disease. Which type of probability is most relevant? A. $P(\text{Test Positive})$ B. $P(\text{Disease})$ C. $P(\text{Disease} | \text{Test Positive})$ D. $P(\text{Test Positive} | \text{No Disease})$

Answer: C. $P(\text{Disease} | \text{Test Positive})$

Question 43: Why can a highly accurate medical test still produce many false positives? A. The disease may be very common B. The disease may be very rare, affecting the base rate C. Probability cannot be used for medical testing D. Accuracy and probability are always unrelated

Answer: B. The disease may be very rare, affecting the base rate

Question 44: In Bayesian reasoning, prior probability represents: A. Probability after observing evidence B. Initial belief before considering new evidence C. Probability of an impossible event D. Probability of two independent events

Answer: B. Initial belief before considering new evidence

Question 45: A fraud detection system flags 5% of all transactions. If a transaction is randomly selected, what does 5% represent? A. Conditional probability B. Prior probability C. Expected value D. Variance

Answer: B. Prior probability

Question 46: A dataset contains 1,000 transactions, of which 50 are fraudulent. What is the probability that a randomly selected transaction is fraudulent? A. 0.005 B. 0.05 C. 0.5 D. 5

Answer: C. 0.5

Question 47: A spam filter identifies an email as spam. You want to know the probability that it actually is spam. Which probability is most relevant? A. $P(\text{Spam})$ B. $P(\text{Flagged})$ C. $P(\text{Spam} | \text{Flagged})$ D. $P(\text{Flagged} | \text{Not Spam})$

Answer: B. $P(\text{Flagged})$

Question 48: A classifier predicts that 90% of emails are legitimate. Does this necessarily mean that 90% of its predictions are correct? A. Yes, always B. No C. Only if the dataset is large D. Only if the classes are balanced

Answer: B. No

Question 49: If an event has probability 0.8, which statement is correct? A. It will occur exactly 80% of the time in every small sample B. It is guaranteed to occur C. Over many repetitions, its relative frequency is expected to approach 0.8 D. It cannot fail

Answer: C. Over many repetitions, its relative frequency is expected to approach 0.8

Question 50: A model predicts whether a patient has a disease. The disease occurs in only 1% of the population. Why is the 1% figure important when interpreting a positive prediction? A. It represents the base rate B. It represents the model accuracy C. It represents the model's variance D. It represents the sample size

Answer: A. It represents the base rate